

Assignment 1 - Alloy

Signatures: $Rentals = [r_0]$ $Cars = [c_0, c_1, c_2]$

Relations: $available = [A_0, A_1, A_2]$ $rented = [R_0, R_1, R_2]$

Requirement #1: All cars are partitioned into the sets available and rented.

Implication: All cars must be either available or rented, but not both.

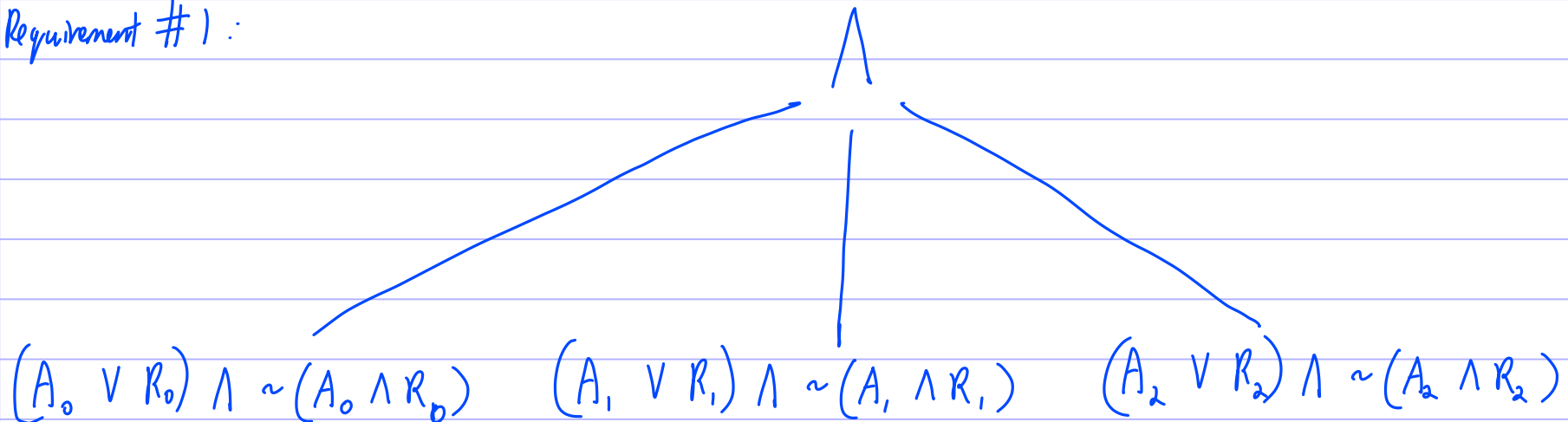
Requirement #2: There is at least one available car.

a) First-order formulae:

Requirement #1: $\forall x \in Car \mid [x \in (available \cup rented)] \vee [x \notin (available \wedge rented)]$

Requirement #2: $\exists x \in Car \mid x \in Available$

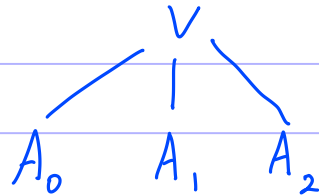
b) Requirement #1:



Final Expression:

$$[(A_0 \vee R_0) \wedge \sim(A_0 \wedge R_0)] \wedge [(A_1 \vee R_1) \wedge \sim(A_1 \wedge R_1)] \wedge [(A_2 \vee R_2) \wedge \sim(A_2 \wedge R_2)]$$

Requirement #2:



Final Expression: $A_0 \vee A_1 \vee A_2$

c) Sample Solution: available = $\begin{bmatrix} 1 & 1 & 0 \end{bmatrix}$
rented = $\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$

d) Alloy Instance: available = $\{c_0, c_1\}$ rented = $\{c_2\}$